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Current Research: *Video/Image Restoration and Recognition; Image Generation; Intelligent Transportation; Industrial Defect Detection; Domain Large Models*

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Education Background

Harbin Institute of Technology, Harbin, Heilongjiang Professor (2025.04- Now)

Northwestern Polytechnical University, Xi'an, China Associate Professor (2021.07- 2025.04)

Work Experience

Harbin Institute of Technology, Computer Application Technology, Harbin, China PhD (2017.03-2021.1)

Harbin University of Science and Technology, Computer Technology, Harbin, China Master (2014.08-2017.03)

Harbin University of Science and Technology, Computer Science and Technology, Harbin, China Undergraduate (2010.09-2014.06)

Representative Publications

Highly cited papers

11. Y. Liang, Q. Liu, **C. Tian**, Y. Xi, R. Chen, D. Yuan, X. Chang, Z. He. Spatial Context-Aware UAV Target Tracking for Consumer Electronics[J]. IEEE Transactions on Consumer Electronics, DOI: 10.1109/TCE.2025.3645235(SCI-IF: 10.9/ ESI Highly Cited papers)

10. **C. Tian**, M. Zheng, B. Li, Y. Zhang, S. Zhang, D. Zhang. Perceptive self-supervised learning network for noisy image watermark removal[J]. IEEE Transactions on Circuits and Systems for Video Technology 34 (8), 7069-7079 (SCI-IF:10.8/ ESI Highly Cited papers)

9. Y. Tian, F. Lin, Y. Li, T. Zhang, Q. Zhang, X. Fu, J. Huang, X. Dai, Y. Wang, **C. Tian**, B. Li, Y. Lv, L. Kovács, F. Wang. UAVs meet LLMs: Overviews and perspectives towards agentic low-altitude mobility[J]. Information Fusion, 2025, 122: 103158. (SCI-IF:17.4/ESI Highly Cited papers)

8. **C. Tian**, C. Zhang, B. Zhang, Z. Li, CLP. Chen, D. Zhang. A Cosine Network for Image Super-Resolution[J]. IEEE Transactions on Image Processing, 2025, 35: 305-316. (SCI-IF:15.3/ESI Highly Cited papers)

7. C. Tian, M. Zheng, W. Zuo, S. Zhang, Y. Zhang, C-W. Lin. A cross Transformer for image denoisng[J]. Information Fusion, 2024, 102: 102043. (SCI-IF:14.8/ESI Highly Cited papers)

6. Q. Zhang, J. Xiao, **C. Tian**, J. Chun-Wei Lin, S. Zhang. A robust deformed convolutional neural network (CNN) for image denoising[J]. CAAI Transactions on Intelligence Technology, 2023, 8(2): 331-342. (SCI-IF:5.1/ESI Highly Cited papers)

5. **C. Tian,** M. Zheng, W. Zuo, B. Zhang, Y. Zhang, D. Zhang. Multi-stage image denoising with the wavelet transform[J]. Pattern Recognition, 134: 109050, 2023. (CCF B/ SCI-IF:8.0/ESI Highly Cited papers/Best Paper Award)

4. **C. Tian**, Y. Xu, B. Zhang, L. Fei, C. Lin. Coarse-to-fine CNN for image super-resolution [J], IEEE Transactions on Multimedia, 23:1489-1502, 2021. (CCF B/SCI-IF: 6.3/ESI Highly Cited papers)

3. **C. Tian**, L. Fei, W. Zheng, Y. Xu, W. Zuo, C. Lin. Deep Learning on Image Denoising: An overview [J], Neural Networks,131,251-275, 2020. (CCF B/SCI-IF:8.05/ ESI Highly Cited papers)

2. **C. Tian**, Y. Xu, Z. Li, W. Zuo, L. Fei, H. Liu. Attention-guided CNN for Image Denoising [J], Neural Networks, 124:117-129, 2020. (CCF B/SCI-IF: 8.05/ESI Highly Cited papers)

1. **C. Tian**, Y. Xu, W. Zuo. Image Denoising Using Deep CNN with Batch Renormalization[J], Neural Networks, 121:461-473, 2020. (CCF B/SCI-IF: 8.05/ ESI Highly Cited papers)

Homepage papers of journals

4. **C. Tian**, Y. Xu, B. Zhang, L. Fei, C. Lin. Coarse-to-fine CNN for image super-resolution [J], IEEE Transactions on Multimedia, 23:1489-1502, 2021. (CCF B/SCI-IF: 6.351/Homepage of IEEE TMM)

3. **C. Tian**, L. Fei, W. Zheng, Y. Xu, W. Zuo, C. Lin. Deep Learning on Image Denoising: An overview [J], Neural Networks,131,251-275, 2020. (CCF B/SCI-IF:8.05/Homepage of NN)

2. **C. Tian**, Y. Xu, Z. Li, W. Zuo, L. Fei, H. Liu. Attention-guided CNN for Image Denoising [J], Neural Networks, 124:117-129, 2020. (CCF B/SCI-IF: 8.05/Homepage of NN)

1. **C. Tian**, Y. Xu, W. Zuo. Image Denoising Using Deep CNN with Batch Renormalization[J], Neural Networks, 121:461-473,

2020. (CCF B/SCI-IF: 8.05/Homepage of NN)

Pattern Recognition Best Paper

1. **C. Tian,** M. Zheng, W. Zuo, B. Zhang, Y. Zhang, D. Zhang. Multi-stage image denoising with the wavelet transform[J]. Pattern Recognition, 134: 109050, 2023. (CCF B/ SCI-IF:8.0/ESI Highly Cited papers/Pattern Recognition Best Paper Award/Hot Paper)

IEEE Signal Processing Society Blog Paper

1. **C. Tian**, Y. Xu, B. Zhang, L. Fei, C. Lin. Coarse-to-fine CNN for image super-resolution [J], IEEE Transactions on Multimedia, 23:1489-1502, 2021. (CCF B/SCI-IF: 6.3/ESI Highly Cited papers)

Premium Award for Best Paper in IET

1. **C. Tian,** Y. Xu, L. Fei, J. Wang, J. Wen, N. Luo. Enhanced CNN for image denoising [J], CAAI Transactions on Intelligence Technology 4 (1), 17-23. (SCI-IF: 5.1/CAAI Excellent paper)

Publications

118. X. Chen, X. Li, W. Sun, W. Zhang, **C. Tian**, W. Liu. Asymmetric Invertible Disentanglement for Image Deraining. IEEE Transactions on Image Processing, 2026. (CCF A, IF: 13.7)

117. L. Zhao, C. Ding, S. Li, **C. Tian**, C. Huang, D. Guan, D. Zhang, Q. Zhu. A Unified Graph Domain Adaptation Framework for Cross-Site Brain Network Analysis. IEEE Transactions on Medical Imaging, 2026. (CCF B, IF: 12.4)

116. J. Fang, S. Yan, X. Yin, Y. Yu, **C. Tian**, J. Liu. Blast: A Stealthy Backdoor Leverage Attack against Cooperative Multi-Agent Deep Reinforcement Learning Based Systems. IEEE Transactions on Cognitive Communications and Networking, 2026. (SCI-IF: 7.0)

115. X. Zhang, Y. Zhang, N. Jin, J. Li, Z. Ren, **C. Tian**, Y. Zhou. Multimodal Deep Learning Approaches in Sports Analytics: A Survey. IEEE Transactions on Consumer Electronics, 2026. (SCI-IF: 9.9)

114. S. Wu, H. Wei, Y. Quan, Y. Xu, **C. Tian***, Q. Miao. ITDRCNN: An Interactive Task-Decoupled RCNN for Enhanced Object Detection. Pattern Recognition, 114190, 2026. (CCF B, SCI-IF: 8.0)

113. **C. Tian**, J. Xie, Q. Zhang, C. Li, W. Zuo, S. Zhang. A Cross-Modal Network for Facial Expression Recognition. IEEE Transactions on Image Processing, 2026. (CCF A, IF: 13.7)

112. Z. Wu, J. Xie, X. Zhang, T. Wang, Y. Zhang, Q. Zhu, **C. Tian***. Adaptive Convolutional Neural Network for Image Super-Resolution. Computational Intelligence, 42(2): e70210, 2026. (CCF C, SCI-IF: 1.9)

111. Y. Zhao, M. Song, Q. Zhang, J. Yang, K. Yoshigoe, **C. Tian**. A Generative Multimodal Network for Facial Expression Recognition. Pattern Recognition, 113518, 2026. (CCF B, SCI-IF: 8.0)

110. J. Fang, Y. Yu, **C. Tian**, L. He, J. Liu. SafaSR: An Arbitrary-Scale Image Super-Resolution Network Based on Multi-Domain Feature Fusion for Enhancing Diverse IoT Vision. IEEE Internet of Things Journal, 2026. (CCF C, SCI-IF: 8.7)

109. J. Lin, X. Zheng, **C. Tian**, X. Lu. A Wavelength-Dependent Haze Distribution Prior Network for Remote Sensing RGB Image Dehazing. IEEE Transactions on Geoscience and Remote Sensing, 2026. (CCF B, SCI-IF: 8.6)

108. C. Li, P. Gong, S. Li, **C. Tian**, Y. Yu, R. Wang, D. Zhang, Q. Zhu. Spatio-Temporal Hypergraph Attention Networks for Brain Disease Analysis. IEEE Transactions on Image Processing, 2026. (CCF A, IF: 13.7)

107. X. Zhang, S. Li, H. Li, P. Gong, J. Shi, J. Shen, **C. Tian**, D. Zhang, Q. Zhu. A Survey on Graph Neural Networks. IEEE Access, 2026. (SCI-IF: 4.2)

106. **C. Tian**, T. Cheng, Y. Ma, Q. Zhu, B. Zhang, D. Zhang. A Heterogeneous CNN With Transformers for Image Denoising. IEEE Transactions on Consumer Electronics, 2026. (SCI-IF: 9.9)

105. J. Xie, K. Zha, P. Liu, **C. Tian***. Operator-Level Reconstruction Method of YOLOv11 for Ascend NPU. Computer Engineering, 52(9): 365–375, 2026.

104. Y. Ma, Y. Wang, K. Feng, J. Xie, Q. Zhu, **C. Tian***. DualExNet: Hybrid Dual-Branch Image Denoising Network. Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2026. (CCF A)

103. Z. Wu, Q. Zhang, X. Sun, Y. Tian, Q. Zhu, **C. Tian***. A Unified Framework for Noisy Image Super-Resolution. ACM Transactions on Autonomous and Adaptive Systems, 2026. (SCI-IF: 2.1)

102. Y. Liang, Q. Liu, **C. Tian**, Y. Xi, R. Chen, D. Yuan, X. Chang, Z. He. Spatial Context-Aware UAV Target Tracking for Consumer Electronics. IEEE Transactions on Consumer Electronics, 2025. (SCI-IF: 9.9)

101. P. Liu, H. Zhang, **C. Tian***, F. Jiang, Y. Zhang, W. Zuo. Tri-Domain Filtering Transformer for CT Metal Artifact Reduction. IEEE Transactions on Radiation and Plasma Medical Sciences, 1–1, 2025. (SCI-IF: 3.5)

100. **C. Tian**, C. Zhang, B. Zhang, Z. Li, C. L. P. Chen, D. Zhang. A Cosine Network for Image Super-Resolution. IEEE Transactions on Image Processing, 2025. (CCF A, IF: 13.7)

99. **C. Tian**, J. Xie, L. Li, W. Zuo, Y. Zhang, D. Zhang. A Perception CNN for Facial Expression Recognition. IEEE Transactions on Image Processing, 2025. (CCF A, IF: 13.7)

98. S. Wu, **C. Tian**, R. Liu, H. Wei, Y. Xu. An Effective Interval Normalization Weighting Method for Accurate Object Detection. IEEE Transactions on Systems, Man, and Cybernetics: Systems, 56(1): 387–399, 2025. (CCF B, IF: 8.7)

97. K. Zeng, Y. Wu, Z. Li, R. Hu, **C. Tian**. DSCIL: Dynamic Selected Contrastive Instance Learning for Weakly Supervised Video Anomaly Detection. Pattern Recognition, 112656, 2025. (CCF B, SCI-IF: 8.0)

96. C. Zheng, **C. Tian**, J. Wen, D. Zhang, Q. Zhu. HeLo: Heterogeneous Multi-Modal Fusion with Label Correlation for Emotion Distribution Learning. Proceedings of the 33rd ACM International Conference on Multimedia, 5519–5527, 2025. (CCF A)

95. **C. Tian**, T. Cheng, Z. Peng, W. Zuo, Y. Tian, Q. Zhang, F. Y. Wang, D. Zhang. A Survey on Deep Learning Fundamentals. Artificial Intelligence Review, 58(12): 381, 2025. (SCI-IF: 10.8)

94. Y. Tian, F. Lin, Y. Li, T. Zhang, Q. Zhang, X. Fu, J. Huang, X. Dai, Y. Wang, et al. UAVs Meet LLMs: Overviews and Perspectives Towards Agentic Low-Altitude Mobility. Information Fusion, 122: 103158, 2025. (SCI-IF: 15.5)

93. C. Song, H. Li, B. Li, J. Wang, **C. Tian**. Distributed Robust Data-Driven Event-Triggered Control for QUAVs under Stochastic Disturbances. Defence Technology, 2025. (SCI-IF: 5.9)

92. W. Xin, Z. Wu, Q. Zhu, T. Bi, B. Li, **C. Tian***. Dynamic Snake Convolution Neural Network for Enhanced Image Super-Resolution. Mathematics, 13(15): 2457, 2025. (SCI-IF: 2.3)

91. X. Chen, Z. Wu, W. Zhang, T. Bi, **C. Tian***. An Omni-Dimensional Dynamic Convolutional Network for Single-Image Super-Resolution Tasks. Mathematics, 13(15): 2388, 2025. (SCI-IF: 2.3)

90. Y. Zhao, Z. Chen, J. He, W. Li, Y. Xiao, **C. Tian**, Y. Zhang. GIP-Stereo: Geometry-Aware Information Propagation Network for Stereo Matching. Knowledge-Based Systems, 114062, 2025. (CCF C, SCI-IF: 8.0)

89. S. Li, Q. Zhu, **C. Tian**, W. Shao, D. Zhang. Interpretable Dynamic Brain Network Analysis with Functional and Structural Priors. IEEE Transactions on Medical Imaging, 44(12): 4878–4889, 2025. (CCF B, IF: 9.8)

88. K. Huang, J. Gui, X. Li, **C. Tian***. Hierarchical Space-Channel Context Correlation for Salient Object Detection. IEEE Transactions on Instrumentation and Measurement, 2025. (SCI-IF: 5.9)

87. **C. Tian**, M. Song, X. Fan, X. Zheng, B. Zhang, D. Zhang. A Tree-Guided CNN for Image Super-Resolution. IEEE Transactions on Consumer Electronics, 71(2): 3631–3640, 2025. (SCI-IF: 9.9)

86. X. Shu, Z. Li, **C. Tian**, X. Chang, D. Yuan. An Active Learning Model Based on Image Similarity for Skin Lesion Segmentation. Neurocomputing, 630: 129690, 2025. (CCF C, SCI-IF: 6.5)

85. **C. Tian**, K. Liu, B. Zhang, Z. Huang, C. W. Lin, D. Zhang. A Dynamic Transformer Network for Vehicle Detection. IEEE Transactions on Consumer Electronics, 71(2): 2387–2394, 2025. (SCI-IF: 9.9)

84. **田春伟**，宋明键，左旺孟，杜博，张艳宁，张师超。卷积神经网络在图像超分辨上的应用。智能系统学报，20(3): 719－749，2025.

83. S. Li, Q. Zhu, **C. Tian**, X. Zhang, W. Shao, J. Wen, D. Zhang. NeuroH-TGL: Neuro-Heterogeneity Guided Temporal Graph Learning Strategy for Brain Disease Diagnosis. Proceedings of the Thirty-Ninth Annual Conference on Neural Information Processing Systems, 2025. (CCF A)

82. W. Xin, Z. Wang, X. Chen, Y. Tang, B. Li, **C. Tian***. 3D Enhanced Residual CNN for Video Super-Resolution Network. Computers, Materials & Continua, 2025. (SCI-IF: 2.4)

81. S. Li, Q. Zhu, **C. Tian**, L. Zhang, B. Shen, C. Zheng, D. Zhang, W. Shao. Spatio-Temporal Evolutionary Graph Learning for Brain Network Analysis Using Medical Imaging. IEEE Transactions on Image Processing, 2025. (CCF A, IF: 13.7)

80. M. Zhu, Y. Gong, D. Gu, **C. Tian***. Improving 3D Object Detection in Neural Radiance Fields With Channel Attention. CAAI Transactions on Intelligence Technology, 2025. (SCI-IF: 7.3)

79. X. Shu, F. Huang, Z. Qiu, **C. Tian**, Q. Liu, D. Yuan. Search Region Updating with Hierarchical Feature Fusion for Accurate Thermal Infrared Tracking. Journal of the Franklin Institute, 361(18): 107332, 2024. (SCI-IF: 7.3)

78. K. Huang, L. Fang, **C. Tian***. Learning to Adapt Using Test-Time Images for Salient Object Detection in Optical Remote Sensing Images. IEEE Transactions on Geoscience and Remote Sensing, 2024. (CCF B, SCI-IF: 7.5)

77. P. Guo, Q. Zhang, **C. Tian**, W. Xue, X. Feng. Digital Human Techniques for Education Reform. Proceedings of the 2024 7th

International Conference on Educational Innovation and Multimedia, 2024. (Conference paper; no journal IF)

76. Q. Meng, **C. Tian***. An Infrared-Visible Image Fusion Network with Multi-Scale Convolutional Attention and Transformer. 2024 International Conference on Cyber-Physical Social Intelligence (ICCSI), 1–6, 2024. (Conference paper; no journal IF)

75. D. Yuan, D. Liao, F. Huang, Z. Qiu, X. Shu, **C. Tian**, Q. Liu. Hierarchical Attention Siamese Network for Thermal Infrared Target Tracking. IEEE Transactions on Instrumentation and Measurement, 73: 1–11, 2024. (SCI-IF: 5.6)

74. L. Zhang, **C. Tian**, H. Song, S. Liu. Multi-Level Dual-Tree Complex Wavelet Network for Low-Dose CT Image Denoising. Computer Engineering, 50(9): 266–275, 2024.

73. M. Zhu, Y. Gong, **C. Tian**, Z. Zhu. A Systematic Survey of Transformer-Based 3D Object Detection for Autonomous Driving: Methods, Challenges and Trends. Drones, 8(8): 412, 2024. (SCI-IF: 4.4)

72. Y. Hu, **C. Tian**, J. Zhang, S. Zhang. Efficient Image Denoising with Heterogeneous Kernel-Based CNN. Neurocomputing, 592: 127799, 2024. (CCF C, SCI-IF: 5.5)

71. **C. Tian**, X. Zhang, Q. Zhang, M. Yang, Z. Ju. Image Super-Resolution via Dynamic Network. CAAI Transactions on Intelligence Technology, 9(4): 837–849, 2024. (SCI-IF: 8.4)

70. Q. Gao, X. Zhang, **C. Tian**, H. Gao, Z. Ju. Human-Centered Robot Vision and Artificial Perception. Frontiers in Robotics and AI, 11: 1406280, 2024. (SCI-IF: 2.9)

69. **C. Tian**, M. Zheng, C. W. Lin, Z. Li, D. Zhang. Heterogeneous Window Transformer for Image Denoising. IEEE Transactions on Systems, Man, and Cybernetics: Systems, 2024. (CCF B, SCI-IF: 8.6)

68. **C. Tian**, H. Gao, P. Wang, B. Zhang. An Enhanced GAN for Image Generation. Computers, Materials & Continua, 80(1), 2024. (SCI-IF: 2.1)

67. Q. Zhang, J. Xiao, S. Zhang, J. C. Lin, **C. Tian**, C. Zhang. Texture-Guided CNN for Image Denoising. Multimedia Tools and Applications, 83(23): 63949–63973, 2024. (CCF C, SCI-IF: 3.0)

66. **C. Tian**, J. Xiao, B. Zhang, W. Zuo, Y. Zhang, C. W. Lin. A Self-Supervised Network for Image Denoising and Watermark Removal. Neural Networks, 174: 106218, 2024. (CCF B, SCI-IF: 13.9)

65. **C. Tian**, M. Zheng, T. Jiao, W. Zuo, Y. Zhang, C. W. Lin. A Self-Supervised CNN for Image Watermark Removal. IEEE Transactions on Circuits and Systems for Video Technology, 34(8): 7566–7576, 2024. (CCF B, SCI-IF: 8.3)

64. K. Huang, N. Li, J. Huang, **C. Tian***. Exploiting Memory-Based Cross-Image Contexts for Salient Object Detection in Optical Remote Sensing Images. IEEE Transactions on Geoscience and Remote Sensing, 2024. (CCF B, SCI-IF: 7.5)

63. Y. Hu, **C. Tian**, C. Zhang, S. Zhang. Efficient Feature Redundancy Reduction for Image Denoising. World Wide Web, 27(2): 20, 2024. (CCF B, SCI-IF: 2.7)

62. **C. Tian**, M. Zheng, W. Zuo, S. Zhang, Y. Zhang, C. W. Lin. A Cross Transformer for Image Denoising. Information Fusion, 102: 102043, 2024. (SCI-IF: 14.8)

61. **C. Tian**, M. Zheng, B. Li, Y. Zhang, S. Zhang, D. Zhang. Perceptive Self-Supervised Learning Network for Noisy Image Watermark Removal. IEEE Transactions on Circuits and Systems for Video Technology, 34(8): 7069–7079, 2024. (CCF B, SCI-IF: 8.3)

60. T. Cheng, T. Bi, W. Ji, **C. Tian***. Graph Convolutional Network for Image Restoration: A Survey. Mathematics, 12(13): 2020, 2024. (SCI-IF: 2.3)

59. **C. Tian**, Y. Zhang, W. Zuo, C. W. Lin, D. Zhang, Y. Yuan. A Heterogeneous Group CNN for Image Super-Resolution. IEEE Transactions on Neural Networks and Learning Systems, 2024. (CCF B, SCI-IF: 10.2)

58. 张鲁，**田春伟**，宋焕生，刘侍刚。多级双树复小波网络用于低剂量 CT 图像去噪。计算机工程，2024.

57. K. Yang, Q. Li, **C. Tian**, H. Zhang, A. Shi, J. Li. DeFort: Deformable Transformer for Visual Tracking. Neural Networks, 2024. (CCF B, SCI-IF: 7.2)

56. **C. Tian**, X. Zhang, X. Liang, B. Li, Y. Sun, S. Zhang. Knowledge Distillation with Fast CNN for License Plate Detection. IEEE Transactions on Intelligent Vehicles, 2023. (SCI-IF: 11.16)

55. X. Zhang, **C. Tian***, Q. Zhang, H. S. Gan, T. Cheng, M. A. H. Ibrahim. Lightweight Image Denoising Network for Multimedia Teaching System. Mathematics, 11(17): 3678, 2023. (SCI-IF: 2.3)

54. K. Huang, **C. Tian***, C. W. Lin. Progressive Context-Aware Dynamic Network for Salient Object Detection in Optical Remote Sensing Images. IEEE Transactions on Geoscience and Remote Sensing, 2023. (CCF B, SCI-IF: 8.6)

53. **C. Tian**, B. Li, J. Zheng, W. Lu, P. Guo, B. Li. Reform of Talent Cultivation Based on Deep Learning Technology. Proceedings of the 2023 2nd International Conference on Educational Innovation and Multimedia, 2023. (Conference paper; no journal IF)

52. Q. Zhang, **C. Tian***, P. Gao. School and Enterprise Cooperation Based on Artificial Intelligence Technology. Proceedings of the 2023 2nd International Conference on Educational Innovation and Multimedia, 2023. (Conference paper; no journal IF)

51. K. Huang, **C. Tian***, Z. Xu, N. Li, J. C. W. Lin. Motion Context Guided Edge-Preserving Network for Video Salient Object Detection. Expert Systems with Applications, 120739, 2023. (CCF C, SCI-IF: 7.5)

50. K. Huang, **C. Tian**, G. Li. Bidirectional Mutual Guidance Transformer for Salient Object Detection in Optical Remote Sensing Images. International Journal of Remote Sensing, 44(13): 4016–4033, 2023. (SCI-IF: 2.6)

49. Q. Zhang, J. Xiao, **C. Tian***, J. Xu, S. Zhang, C. W. Lin. A Parallel and Serial Denoising Network. Expert Systems with Applications, 120628, 2023. (CCF C, SCI-IF: 7.5)

48. **C. Tian**, M. Zheng, W. Zuo, B. Zhang, Y. Zhang, D. Zhang. Multi-Stage Image Denoising with the Wavelet Transform. Pattern Recognition, 134: 109050, 2023. (CCF B, SCI-IF: 8.0)

47. L. Fu, B. Shi, L. Sun, J. Zeng, D. Chen, H. Zhao, **C. Tian**. An Improved U-Net for Watermark Removal. Electronics, 11(22): 3760, 2022. (SCI-IF: 2.9)

46. K. Huang, **C. Tian***, J. Su, J. C. Lin. Transformer-Based Cross Reference Network for Video Salient Object Detection. Pattern Recognition Letters, 2022. (CCF C, SCI-IF: 3.3)

45. **C. Tian**, Y. Yuan, S. Zhang, C. W. Lin, W. Zuo, D. Zhang. Image Super-Resolution with an Enhanced Group Convolutional Neural Network. Neural Networks, 153: 373–385, 2022. (CCF B, SCI-IF: 7.2)

44. M. Zheng, K. Zhi, J. Zeng, **C. Tian**, L. You. A Hybrid CNN for Image Denoising. Journal of Artificial Intelligence and Technology, 2022.

43. J. Song, J. Xiao, **C. Tian**, Y. Hu, L. You, S. Zhang. A Dual CNN for Image Super-Resolution. Electronics, 11(5): 757, 2022. (SCI, IF: 2.395)

42. J. Xiao, J. Xu, **C. Tian***, P. Han, L. You, S. Zhang. A Serial Attention Frame for Multi-Label Waste Bottle Classification. Applied Sciences, 12(3), 2022. (SCI-IF: 2.679)

41. L. Zhu, C. Zhang, J. Song, S. Zhang, **C. Tian**, X. Zhu. Deep Multigraph Hierarchical Enhanced Semantic Representation for Cross-Modal Retrieval. IEEE MultiMedia, 29(3): 17–26, 2022. (SCI, IF: 5.622)

40. Y. Hu, H. Wang, C. Jin, B. Li, **C. Tian***. A Diffusion Model with an FFT for Image Inpainting. Journal of Cyber Physical Social Intelligence, 60–69, 2022.

39. Q. Zhang, J. Xiao, **C. Tian***, J. Chun-Wei Lin, S. Zhang. A Robust Deformed Convolutional Neural Network (CNN) for Image Denoising. CAAI Transactions on Intelligence Technology, 2022. (SCI-IF: 7.3)

38. M. Zheng, J. Xu, Y. Shen, **C. Tian***, J. Li, L. Fei, M. Zong, X. Liu. Attention-Based CNNs for Image Classification: A Survey. Journal of Physics: Conference Series, 2171(1): 012068, 2022. (EI)

37. Y. Deng, L. Fei, J. Wen, W. Jia, G. Zhao, **C. Tian**, T. Ke. Joint Multi-Feature Learning for Facial Age Estimation. Asian Conference on Pattern Recognition, 513–524, 2021. (EI)

36. T. Chunwei, Y. Yixuan. An Unsupervised Neural Network for Image Denoising, 2021

35. T. Wang, Y. Liu, Q. Yin, J. Geng, J. Chen, X. Yin, Y. Wang, X. Shang, **C. Tian***, et al. Enhancing Discoveries of Molecular QTL Studies with Small Sample Size Using Summary Statistic Imputation. Briefings in Bioinformatics, 2021. (CCF B, SCI-IF: 11.622)

34. **C. Tian**, Y. Xu, W. Zuo, B. Du, C. W. Lin, D. Zhang. Designing and Training of a Dual CNN for Image Denoising. Knowledge-Based Systems, 226: 106949, 2021. (CCF C/SCI-IF: 8.038)

33. H. Zhang, **C. Tian***, L. You, Z. Li, M. Zong, K. Huang. Design and Implementation on Matching Between Music and Color. Multimedia Tools and Applications, 2021. (CCF C, SCI-IF: 3.3)

32. L. Fei, B. Zhang, **C. Tian**, S. Teng, J. Wen. Jointly Learning Multi-Instance Hand-Based Biometric Descriptor. Information Sciences, 562: 1–12, 2021. (CCF B/SCI-IF: 6.795)

31. J. Qin, L. Fei, J. Zhu, J. Wen, **C. Tian**, S. Wu. Scalable Discriminative Discrete Hashing for Large-Scale Cross-Modal Retrieval. ICASSP 2021, 4330–4334, 2021. (CCF B/EI)

30. **C. Tian**, Y. Xu, W. Zuo, C. W. Lin, D. Zhang. Asymmetric CNN for Image Super-Resolution. IEEE Transactions on Systems, Man, and Cybernetics: Systems, 52(6): 3718–3730, 2021. (CCF B, SCI-IF: 13.451)

29. L. Fei, B. Zhang, Y. Xu, **C. Tian**, I. Rida, D. Zhang. Jointly Heterogeneous Palmprint Discriminant Feature Learning. IEEE Transactions on Neural Networks and Learning Systems, 33(9): 4979–4990, 2021. (CCF B/SCI-IF: 10.451)

28. **C. Tian**, Y. Xu, W. Zuo, B. Du, L. Fei, C. W. Lin. Coarse-to-Fine CNN for Image Super-Resolution. IEEE Transactions on Multimedia, 23: 1489–1502, 2021. (CCF B/SCI-IF: 6.351)

27. **C. Tian**, R. Zhuge, Z. Wu, Y. Xu, W. Zuo, C. Chen, C. W. Lin. Lightweight Image Super-Resolution with Enhanced CNN. Knowledge-Based Systems, 2020. (CCF C/SCI-IF: 8.038)

26. **C. Tian**, L. Fei, W. Zheng, Y. Xu, W. Zuo, C. W. Lin. Deep Learning on Image Denoising: An Overview. Neural Networks,

131: 251–275, 2020. (CCF B/SCI-IF: 8.05)

25. L. Fei, S. Teng, J. Wen, **C. Tian**, D. Liu, W. Zhang. Jointly Learning Compact Multi-Instance FKP Descriptor. International Joint Conference on Biometrics (IJCB), 2020. (EI)

24. L. Fei, B. Zhang, J. Wen, **C. Tian**, P. Liu, S. Zhao. Joint Learning Multiple Curvature Descriptor for 3D Palmprint Recognition. International Conference on Pattern Recognition (ICPR), 2020. (EI)

23. **C. Tian**, Y. Xu, Z. Li, W. Zuo, L. Fei, H. Liu. Attention-Guided CNN for Image Denoising. Neural Networks, 124: 117–129, 2020. (CCF B/SCI-IF: 8.05)

22. **C. Tian**, Y. Xu, W. Zuo. Image Denoising Using Deep CNN with Batch Renormalization. Neural Networks, 121: 461–473, 2020. (CCF B/SCI-IF: 8.05)

21. Y. Xu, Z. Li, **C. Tian**, J. Yang. Multiple Vector Representations of Images and Robust Dictionary Learning. Pattern Recognition Letters, 128: 131–136, 2019. (CCF C, SCI-IF: 3.756)

20. L. Fei, B. Zhang, S. Teng, A. Zeng, **C. Tian**, W. Zhang. Learning Discriminative Finger-Knuckle-Print Descriptor. ICASSP 2019, 2137–2141, 2019. (CCF B, EI)

19. **C. Tian**, Y. Xu, L. Fei, J. Wang, J. Wen, N. Luo. Enhanced CNN for Image Denoising. CAAI Transactions on Intelligence Technology, 4(1): 17–23, 2019. (SCI)

18. **C. Tian**, Y. Xu, L. Fei, K. Yan. Deep Learning for Image Denoising: A Survey. International Conference on Genetic and Evolutionary Computing, 563–572, 2018. (EI)

17. J. Wen, X. Fang, Y. Xu, **C. Tian**, L. Fei. Low-Rank Representation with Adaptive Graph Regularization. Neural Networks, 108: 83–96, 2018. (CCF B/SCI-IF: 8.05)

16. Y. Qin, **C. Tian**. Weighted Feature Space Representation with Kernel for Image Classification. Arabian Journal for Science and Engineering, 43(12): 7113–7125, 2018. (SCI-IF: 2.334)

15. **C. Tian***, Q. Zhang, G. Sun, Z. Song, S. Li. FFT Consolidated Sparse and Collaborative Representation for Image Classification. Arabian Journal for Science and Engineering, 43(2): 741–758, 2018. (SCI-IF: 2.334)

14. J. Dong, **C. Tian**, Y. Xu. Face Liveness Detection Using Color Gradient Features. 2017 International Conference on Security, Pattern Analysis, and Cybernetics, 377–382, 2017. (EI)

13. L. Fei, S. Teng, J. Wu, Y. Xu, J. Wen, **C. Tian**. Combining Enhanced Competitive Code with Compacted ST for 3D Palmprint Recognition. 4th IAPR Asian Conference on Pattern Recognition, 512–517, 2017. (EI)

12. **C. Tian**, Q. Zhang, G. Sun. Multiple Methods for WeChat Identification. The 6th International Conference on Advanced Design and Manufacturing Engineering, 2017. (Conference paper; no journal IF)

11. **C. Tian**, G. Sun, Q. Zhang, W. Wang, T. Chen, Y. Sun. Integrating Sparse and Collaborative Representation Classifications for Image Classification. International Journal of Image and Graphics, 17(2): 1750007, 2017.

10. **C. Tian**, Q. Zhang, J. Zhang, G. Sun, Y. Sun. 2D-PCA Representation and Sparse Representation for Image Recognition. Journal of Computational and Theoretical Nanoscience, 14(1): 829–834, 2017.

9. **C. Tian**, G. L. Sun, S. X. Zhu, Q. Zhang. WeChat Identification Based on Network Traffic Characteristics. Information Science and Electronic Engineering: Proceedings of the 3rd International Conference, 2016. (EI)

8. **C. Tian**, Q. Zhang, G. Sun, Y. Sun, Y. Qin. Weighted Nearest Neighbor Algorithm and Collaborative Representation for Image Classification. Journal of Computational and Theoretical Nanoscience, 13(12): 9065–9070, 2016.

7. **C. Tian**, Q. Zhang, G. Sun, Y. Sun. Linear Discriminant Analysis Representation and CRC Representation for Image Classification. 2016 2nd IEEE International Conference on Computer and Communications, 755–760, 2016. (EI)

6. Z. Hong, G. Xiaoshuang, M. Obadi, Z. Ruirui, T. Jinrui, **C. Tian**, et al. Research and Implementation of RGB-LED Mixed Light Matching Along with Music Playing. International Journal of Grid and Distributed Computing, 9(4), 2016.

5. W. Wang, Z. Yu, **C. Tian**. An Analysis Algorithm of Pipe Burst in Water Supply Network and Its Implementation Based on GIS Components. Journal of Harbin University of Science & Technology, 20(4), 2015.

4. Z. Hong, Z. Ruirui, **C. Tian**. Design and Implementation of the Music Player Based on Interface Chip CH378. 2015 Seventh International Conference on Advanced Communication and Networking, 2015. (EI)

3. D. Li, **C. Tian**, B. Li, G. Sun, J. Kang. Network Traffic Classification Based on Subspace Clustering-Based Method. Journal of Harbin University of Science and Technology, 20(2): 63–68, 2015.

2. Z. Hong, T. **C. Tian**, G. Lin, X. Li, L. Lu. Design and Implementation of Mixed Light Effect Based on the Audio Signal Spectrum Analysis. Journal of Harbin University of Science & Technology, 18(6): 47, 2013.

1. Z. Hong, Z. Ruirui, **C. Tian**. Implementation of a Wireless Controlled Device with RGB Driving Based on FFT. International Journal of Multimedia and Ubiquitous Engineering, 8(6): 151–158, 2013.

Social Activities

British Computer Society Fellow (BCS Fellow);

Stanford University World's Top 2% Scientists (2022–2026); Stanford University World's Top 2% Scientists: Career-long Impact List

Editorial board member: IEEE TIP **(**SCI-IF:15.3**)** Associate Editor; PR **(**SCI-IF:9.1**)** Associate Editor; IEEE TCE **(**SCI-IF:9.9**)** Associate Editor; Computational Intelligence **(**SCI-IF:1.9**)** Area Editor;

Reviewer: IEEE TPAMI、IEEE TIP、IEEE TNNLS、IEEE TMM、IEEE TCSVT、IEEE TGRS、IEEE TIFS、IJCV、PR、CVPR、AAAI、IJCAI、ACM MM、WWW、ICML、ICLR、NeurIPS;

Delivered keynote speeches at the 2025 International Conference on Artificial Intelligence and Education, the 2023 International Conference on Cyber-physical Social Intelligence, the 2023 China-Europe International Symposium on Software Engineering Education (CEISEE), and the 2025 China-Europe International Symposium on Software Engineering Education (CEISEE).